

Kristopher Kirkwood

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EDUCATION

Engineering Physics Bachelor of Applied Science

University of British Columbia
2021 - 2026

An interdisciplinary blend of software, electrical, and mechanical engineering, emphasizing math and physics skills for project development.

SKILLS

Software

Python	C/C++
Java	HTML/CSS
Linux	Git
KiCad	Arduino
Labjack	Onshape
Solidworks	Fusion 360
MatLab	Cura
LaTeX	Excel / Word

Manufacturing/Prototyping

3D Printing	Laser Cutting
Soldering	CNC
Mill	Lathe
Band Saw	Composites
Waterjet	Oscilloscope

Personal

Hard Working	Creative
Dependable	Passionate

INTERESTS

Intramurals, Hiking, Cycling
Paddleboarding, Camping,
Running, Woodworking

AWARDS

AP Scholar with Distinction
16 Scholarships including:
EGBC Scholarship
Academic Achievement Award
BC Principals Scholarship
District Athletics Award
Donald J. Evans Scholarship in Engineering
UBC Trek Scholarship (2x)

CONFERENCES

USA National Student Leadership Summit

NASA Student Spaceflight Experiments Program

WORK EXPERIENCE

Rocket Lab

May 2025 – Aug 2025

Avionics Intern – Neutron Avionics Systems

- Designed, manufactured, and installed electrical harnesses for avionics flight hardware.
- Sourced and purchased aerospace-grade connectors and cables for Neutron rocket.

D-Wave Systems Inc.

May 2024 – Dec 2024

Mechatronics - Lab Operations Engineering Co-op

- Fully automated a magnetic screening system which utilizes superconducting quantum interference devices to precisely detect nano-tesla magnetic fields.
- Designed and built custom circuits and PCBs for sensor integration with a LabJack.
- Developed firmware to control a motor based on pressure and light sensor inputs.
- Coded a user interface that provides real-time actuator positioning and control.
- Operated cryogenic equipment for quantum computing experiments and validated processor viability through resistive testing in a LISP and emacs environment.

DarkVision Technologies

Jan 2023 – Apr 2023

Mechanical and Mechatronics Engineering Co-op

- Designed and fabricated apparatus for testing and assembly of downhole, and pipeline ultrasonic imaging tools. Conducted tests on downhole tools to diagnose failure modes and maximum tool stresses.
- Designed and built a prototype odometer to monitor tool speed and positioning and laser-cut a jig to test circuit boards in a high salinity solution.
- Created CAD files and manufacturing drawings to GD&T standards for outsourced parts and made custom parts on Tormach CNC machines.

TECHNICAL EXPERIENCE

Electromagnetically Powered Autonomous Sphere

Sept 2024 – Present

- Creating a sphere that rolls using forces generated by custom electromagnets.
- Designing a chopper drive circuit to create a precise current control feedback loop.
- Developing control systems to precisely manipulate magnetic forces.

UBC Engineering Physics Robot Summer

July 2023 – Aug 2023

- Worked in a team to design and build an autonomous robot that accurately followed a tape track while picking up and rejecting blocks at a high speed.
- Designed, built, and tested electrical circuits to distribute power and isolate sensors from batteries, and detect a tape line and magnetic blocks.
- Used function generators, oscilloscopes, and multimeters to troubleshoot circuits that contained capacitors, resistors, diodes, transistors, and chips.
- Programmed a Blue-Pill to output PWM signals to motors given sensor inputs.

Machine Learning and Neural Networks

Sept 2023 – Dec 2023

- Used OpenCV and TensorFlow within Python to extract data from and control a virtual robot within a simulated ROS / Gazebo environment in Linux.
- Wrote a feedback control algorithm to keep a robot centered on a windy road.
- Implemented CV algorithms to track objects and control a robot from a camera.
- Designed a SIFT algorithm to identify objects in real-time from a webcam input.
- Used neural networks and machine learning to read signs in a video feed.

Diving Score Tracker App - [capkirk.pythonanywhere.com](https://github.com/capkirk/pythonanywhere.com)

June 2024

- Developed an app in Python to track and tabulate scores for diving competitions.
- UI improves efficiency and eliminates opportunities for calculation errors.
- Creates a PDF of results, separated by division and score breakdowns for athletes.

UBC Supermileage Powertrain Division

Sept 2021 – May 2022

- Designed gearbox, motor, and baseplate mounts using SolidWorks.
- Reduced weight by 25% through use of FEA and waterjet manufacturing.
- Designed and 3D printed test mounts for a hydrogen tank on the fuel cell vehicle.